ThermoGo: Medicines Monitoring System

IoT-based environmental monitoring for safe medicine storage

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OBJECTIVE

- Monitor medicine storage temperature and humidity.
- Send readings to the cloud in real time.
- Display data on a mobile app.
- Alert users when conditions are unsafe.
- Build a low-power and user-friendly IoT prototype.

INTRODUCTION

- Some medicines are sensitive to temperature and humidity.
- Poor storage conditions can reduce medicine quality.
- ThermoGo helps users monitor storage conditions remotely using an IoT device and mobile app.

PROBLEM STATEMENT

- Professional monitoring systems are expensive.
- Simple devices often lack cloud storage and configurable alerts.
- Existing IoT platforms can be too technical for normal users.
- Users need a simple, affordable, and connected monitoring solution.

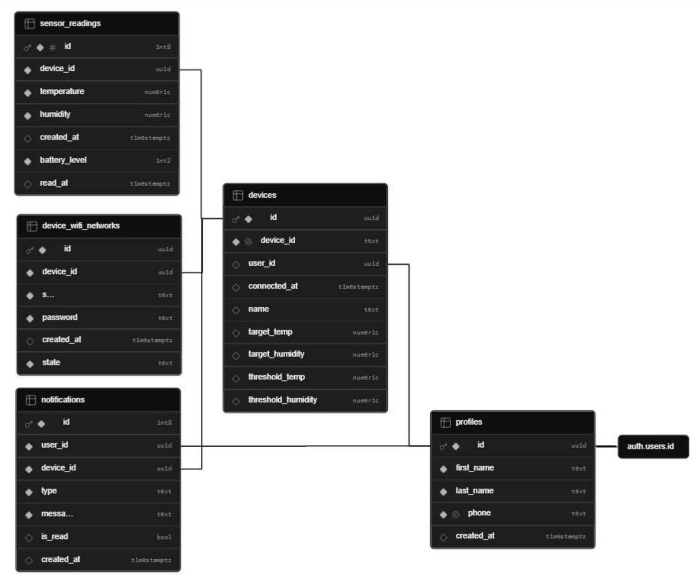
PROPOSED SOLUTION

- ESP32-based monitoring device.
- BME280 temperature and humidity sensor.
- Supabase cloud backend.
- React Native mobile app.
- Push notifications and AI insights.

USE CASE / IMPACT

- Supports safe storage of medicines and sensitive products.
- Reduces manual checking with real-time cloud monitoring.
- Helps users react quickly when temperature or humidity becomes unsafe.
- Provides a portable and low-power prototype for controlled storage.

Database Schema



Users, devices, readings, WiFi networks, and notifications.

BACKEND HIGHLIGHTS

- Supabase PostgreSQL database
- Secure user authentication
- Row-Level Security for privacy
- Time-series sensor data storage
- Fast queries using device/date indexes

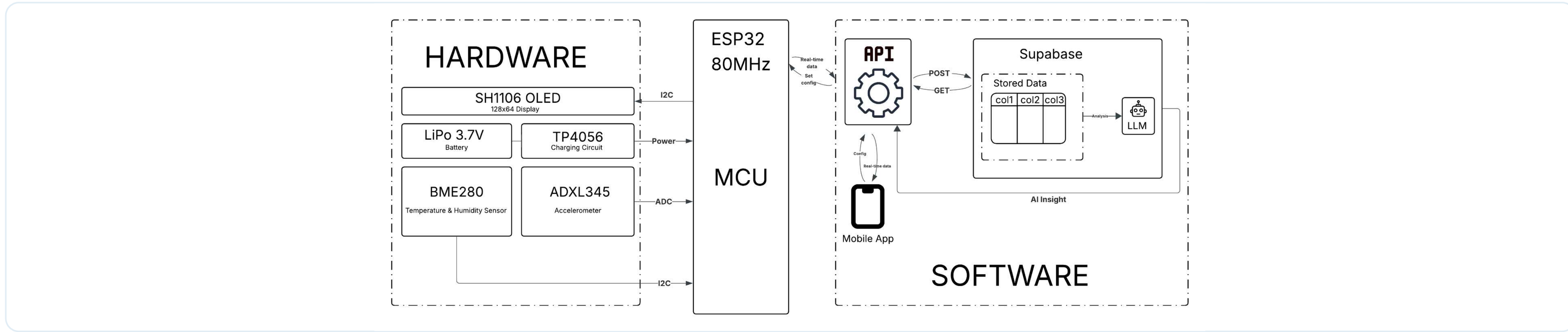
DATA FLOW & ALERTS

- ESP32 sends readings to Supabase through secure API requests.
- Readings are stored with device ID, timestamp, temperature, humidity, and battery level.
- Threshold checks trigger alerts when storage conditions become unsafe.
- Mobile app shows live values, charts, alerts, and device status.

MATERIALS / TECHNOLOGIES

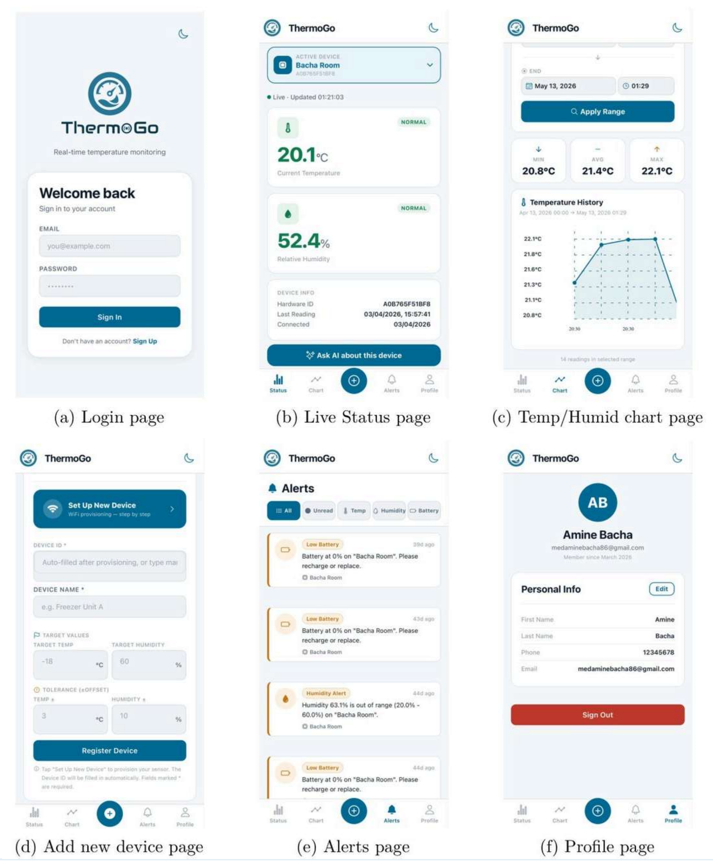
- **Hardware:** ESP32, BME280, ADXL345, LiPo battery, TP4056, MT3608, custom PCB, 3D enclosure.
- **Software:** Arduino / PlatformIO, React Native, Expo, VS Code.
- **Cloud:** Supabase, Supabase Auth, Edge Functions, Gemini API.
- **Design:** SolidWorks, Altium Designer.

SYSTEM ARCHITECTURE / WORKFLOW



Detailed interaction between hardware, ESP32 firmware, API, Supabase backend, AI insight layer, and the mobile app.

Mobile Application Screens

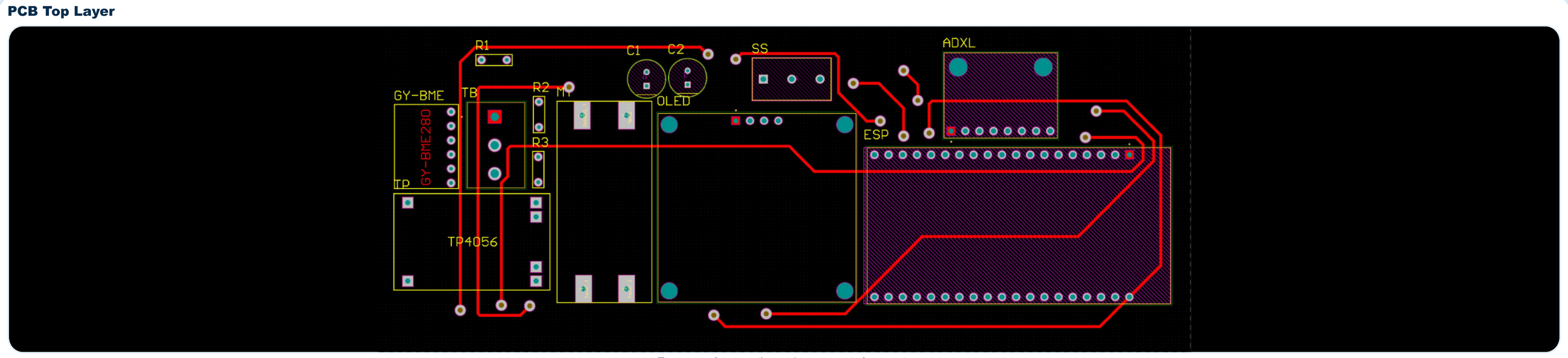


Login, live status, chart, add device, alerts, and profile screens.

3D Product Design

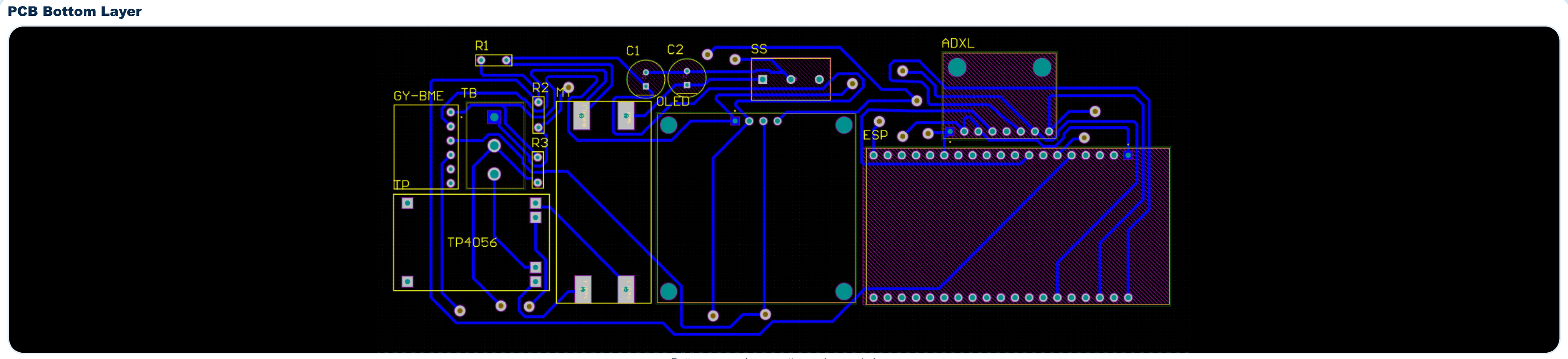
Exploded 3D view of the ThermoGo enclosure and internal components.

PCB Top Layer



Top copper layer routing and component placement.

PCB Bottom Layer



Bottom copper layer routing and ground plane.

METHODOLOGY

- Agile development with short sprints.
- Hardware and software design.
- Firmware, backend, and app implementation.
- System integration.
- Unit, integration, and field testing.

KEY FEATURES

- Real-time monitoring and cloud storage.
- Configurable thresholds and push notifications.
- Historical charts and battery monitoring.
- Adaptive sleep for power saving.
- AI-powered environmental insights.

AI FEATURES

- Gemini-powered environmental insights.
- OLED status messages for unsafe readings.
- Ask AI screen for sensor history questions.
- Personalized feedback using thresholds.

POWER OPTIMIZATION

- ESP32 deep sleep reduces energy use.
- Adaptive sleep changes sampling interval.
- OLED turns off before deep sleep.
- Local buffering reduces WiFi usage.

VALIDATION / TESTING

- 30-day indoor field test.
- End-to-end data flow tested.
- Sensor readings matched reference values.
- Cloud delivery reliability: 98.5%.
- App refresh time: 0.8 sec.

RESULTS

- Complete working IoT system developed and integrated.
- Sensor, WiFi, cloud, storage, and app tests passed.
- Mobile app displays readings, charts, alerts, and device information.

98.5%	±0.3°C	99.8%
Successful cloud deliveries	Temperature validation error	Cloud uptime

CONCLUSION

ThermoGo provides a practical IoT solution for monitoring medicine storage conditions. It combines an ESP32 device, cloud backend, and mobile app to help users track conditions and receive alerts when storage limits are exceeded.

FUTURE WORK

- Solar charging and cellular connectivity.
- Smaller PCB and enclosure.
- Predictive AI analytics.
- Data export.

SCAN & EXPLORE

Use the QR codes below to access the mobile app preview and the ThermoGo brand website.

Mobile App
thermogo.netlify.app

Brand Website
thermogo-brand.vercel.app